

DTS407TC A2: Adversarial Debate Argument Generator

Group ID: _____

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1 Motivation

Debate preparation requires more than producing plausible arguments. A useful tool must expose likely counterarguments, repair weak reasoning, and connect claims to evidence. Standard single-pass prompting often creates fluent but under-tested arguments. This project therefore implements a domain-specific large model system for debate argument generation. Given a motion and a target side, the system returns three debate-ready arguments and compares a direct generation strategy with an adversarial multi-agent strategy.

The task is narrow, realistic, and suitable for a large-model-based system. It involves controlled generation, retrieval from public sources, role-conditioned reasoning, and evaluation of quality and efficiency. The intended users are debate students, presentation teams, and learners practising critical thinking.

2 System Design

The system has a FastAPI backend, a React frontend, a Tavily search wrapper, and an Ollama local model client using `qwen3.5:4B`. The frontend supports English and Chinese, topic selection, target-side selection, live generation, result export, and side-by-side comparison.

Two strategies are implemented under the same task setting:

- **Single Agent:** one prompt uses `/no_think`, search sources, and directly generates three arguments.
- **Adversarial Debate:** four debaters are prompted with `/think`: Pro Logic, Pro Evidence, Con Logic, and Con Evidence. They first produce constructive speeches, then rebut opposing speeches. A neutral synthesis agent extracts the strongest three arguments for the requested side. The implementation includes an `OLLAMA_ENABLE_THINKING` switch because the small local model can spend substantial time in the hidden reasoning channel during live demos.

The adversarial design is intended to increase robustness. It forces arguments to survive attacks from the other side before being shown to the user. The system strips hidden thinking blocks from all outputs before UI display or export.

3 Dataset and Knowledge Sources

The project includes a bilingual dataset of 30 debate motions across five domains: technology ethics, education, social policy, environment, and economy. Each item contains an ID, domain, English motion, Chinese motion, pro label, con label, and source note. The dataset was selected to ensure that both sides have plausible arguments and that public evidence is available through web search.

Tavily is used as the external search source. Search results are cached locally to make demonstrations reproducible and reduce repeated API calls. Each generated result stores source titles, URLs, and snippets.

4 Experiment Design

The core experiment compares two modelling strategies under controlled conditions. Both strategies receive the same motion, target side, language, model, and search tool. The independent variable is the generation architecture: direct single-agent prompting versus adversarial multi-agent debate. The dependent variables are output quality, diversity, evidence use, response time, and approximate token cost.

The lightweight automatic evaluation records response time, token estimate, number of sources, argument count, and character n-gram diversity. The exported CSV includes manual scoring columns so human evaluation can judge persuasiveness, relevance, and practical debate usefulness.

5 Evaluation Methodology and Results

Quantitative results should be generated with:

```
uv run python evaluation/run_eval.py --limit 10 --language en --side pro
```

After running the evaluation, paste the resulting table here.

Strategy	Avg. time	Avg. tokens	Avg. sources	Diversity
Single Agent	TBD	TBD	TBD	TBD
Adversarial Debate	TBD	TBD	TBD	TBD

Table 1: Lightweight automatic evaluation results.

Qualitative analysis should compare 2-3 representative motions. The expected pattern is that the single-agent strategy is faster and cheaper, while the adversarial strategy gives more resilient arguments because it includes rebuttal-aware synthesis.

6 Critical Analysis of Ethical Issues

The system can strengthen one-sided persuasion, which creates misuse risks in political communication, marketing, and online manipulation. It may also reproduce bias from web sources and from the base model. Search snippets can be incomplete or misleading, so the system exposes source links and does not claim that generated evidence is fully verified.

The system mitigates these risks by showing both sides in the internal transcript, preserving sources, supporting manual evaluation, and framing outputs as debate preparation rather than factual authority. Privacy risks are limited because the system runs the model locally and does not require user accounts, but search queries are still sent to Tavily.

7 Conclusion

This project demonstrates that architectural choices can change large-model behaviour. A multi-agent adversarial workflow costs more time and tokens, but it can produce arguments that better anticipate objections. The system therefore provides a practical example of comparative reasoning in domain-specific large model design.